Rafael Enrique Velásquez Rojas

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EDUCATION

Jan 2025 – Expected Apr. 2027	**M.Sc. in Biomedical Engineering** Universidad de los Andes, Bogotá, Colombia M.Sc. thesis in progress – advisor Prof. Pablo Arbeláez.
Jan 2020 – Apr. 2024	**B.Sc. in Physics** Universidad de los Andes, Bogotá, Colombia Thesis on *information transmission in genetic networks* – advisor Prof. Juan Manuel Pedraza.
Aug 2020 – Oct. 2026	**B.Sc. in Biomedical Engineering** Universidad de los Andes, Bogotá, Colombia Design project: *BiliScan* – non-invasive, skin-tone-independent qualitative detection of neonatal jaundice using digital image processing and deep learning.

PUBLICATIONS & PREPRINTS

[1] **Rafael Velásquez**[*], Juan Manuel Pedraza. *Cost-information trade-offs partially rationalize the feed-forward loop census*. Submitted to **Physical Review E.** [code] ([*]corresponding author)

[2] **Rafael Velásquez**, Pablo Arbeláez. *Predicting When Cheap Simulators Break Neural Inference of Bursty Gene Expression*. **LatinX in AI (LXAI) Workshop, NeurIPS 2026.**

[3] Sergio Rincón[*], Gabriel González[*], Nicolás Andrade, **Rafael Velásquez**, Sebastián Ojeda, Juanita Puentes, Nicolás Aparicio, Paula Cárdenas, Pablo Arbeláez. *Quantitative Assessment of Crystal Structure Prediction*. **NeurIPS 2026, Evaluations & Datasets Track.** [paper] ([*]equal contribution)

[4] **Rafael Velásquez**, Esther Puyol-Antón, Pablo Arbeláez. *Beyond Volume Overlap: Surface Matching for Topology-Aware Coronary Artery Segmentation*. **STACOM Workshop, MICCAI 2026.** [paper]

[5] Sebastián Ojeda[*], **Rafael Velásquez**, Nicolás Aparicio, Juanita Puentes, Paula Cárdenas, Nicolás Andrade, Gabriel González, Sergio Rincón, Carolina Muñoz-Camargo, Pablo Arbeláez. *A Standardized Benchmark for Multilabel Antimicrobial Peptide Classification*. **NeurIPS 2025, Datasets & Benchmarks Track.** [paper] ([*]corresponding author)

RESEARCH EXPERIENCE

2024 – Present **Drug Discovery as Next Geometric Token Prediction**
Prof. Pablo Arbeláez, Universidad de los Andes

- Designed a geometric tokenization that discretizes 3D protein and ligand structures, and trained an autoregressive model to predict binding affinity by next-token prediction, improving MSE by about 20% on the BindingDB benchmark.
- Reframed structure-based binding prediction as autoregressive generation over geometric tokens, transferring sequence-model machinery to 3D molecular data.

2025 – Present **Topology-Aware Segmentation of Tubular Structures**
Prof. Pablo Arbeláez, Universidad de los Andes; E. Puyol-Antón, King's College London

- Developed topology-aware training objectives for 3D segmentation of thin tubular structures, including a bipartite surface-matching metric and a surface-distance loss with explicit creation and removal terms that penalize broken or spurious branches invisible to volumetric overlap.
- Used centerline and signed-distance-function representations to make supervision sensitive to connectivity rather than voxel overlap, and weighted errors by $1/r^2$ to protect small distal vessels.

– Benchmarked nnU-Net, SwinUNETR and NexToU on ImageCAS and ASOCA, and am extending the method to a multi-dataset benchmark (Medical Tubathlon) toward a generalist model for tubular structures across body regions.

2023 – 2026 **Information in Gene-Regulatory Networks**

Prof. Juan Manuel Pedraza, Universidad de los Andes

– Built stochastic models (Gillespie and Langevin) of transcriptional regulatory motifs as noisy information channels and carried out the full theoretical and computational analysis, deriving an information-versus-cost trade-off that helps account for the observed prevalence of feed-forward loops. Submitted to Physical Review E.
– Developed a working command of stochasticity in biological systems, treating cell-to-cell variability in gene-regulatory networks as a quantifiable, functional signal rather than mere noise.

2024 – 2026 **Flow Matching for Crystal Structure Prediction**

Prof. Pablo Arbeláez, Universidad de los Andes

– Developed OPAL-CSP, a flow-matching generative model that samples a crystal's lattice and fractional atomic coordinates from its chemical composition, built on an SE(3)-equivariant graph transformer with separate velocity-field heads for lattice and coordinates and a wrapped coordinate path that respects periodic boundary conditions.
– Co-designed OPAL-Bench, a polymorph-aware evaluation framework with species-aware SE(3) alignment and threshold-free geometric metrics, and added physical-plausibility metrics (formation energy, SOAP, XRD) that reveal geometric accuracy does not imply physical validity. Accepted at NeurIPS 2026 (Evaluations and Datasets Track).

2025 **Antimicrobial Peptide Classification (ESCAPE)**

Prof. Pablo Arbeláez, Universidad de los Andes

– Compiled and harmonized numerous public peptide databases into a standardized multilabel benchmark for antimicrobial activity, the project's central contribution.
– Built a dual-encoder architecture that processes amino-acid sequences with a text transformer and 3D-derived distance maps with a vision transformer, fuses the two representations by cross-attention through a class token, and predicts multilabel activity with an MLP head.

2023 – 2025 **Stochastic Modelling of Ageing**

Prof. A. Rutenberg, Dalhousie University (Mitacs Globalink)

– Modelled ageing as stochastic damage and repair dynamics on a network of health attributes, during a Mitacs Globalink internship.

TALKS, POSTERS & CONFERENCES (*SCHEDULED)

Dec 2025 [1] Poster, *A Standardized Benchmark for Multilabel Antimicrobial Peptide Classification*, **NeurIPS 2025** (Datasets & Benchmarks Track), San Diego, USA.

Sep 2026 [2] Poster, *Beyond Volume Overlap: Surface Matching for Topology-Aware Coronary Artery Segmentation*, **STACOM Workshop, MICCAI 2026**, Strasbourg, France.

Dec 2026 [3*] Poster, *Quantitative Assessment of Crystal Structure Prediction*, **NeurIPS 2026** (Evaluations & Datasets Track), Sydney, Australia.

[4*] Poster, *Predicting When Cheap Simulators Break Neural Inference of Bursty Gene Expression*, **LXAI Workshop, NeurIPS 2026**, Sydney, Australia.

TECHNICAL SKILLS

Machine learning Flow matching, diffusion, transformers, Large Language Models (LLMs), equivariant and graph neural networks, multimodal and autoregressive models

Statistical modelling statistical physics, stochastic simulation (Gillespie, Langevin), information theory, simulation-based inference

Biomedical data 3D medical image segmentation, gene-regulatory network modelling, molecular and structural data

Tools Python, PyTorch, Linux, HPC, Git

SCHOLARSHIPS & AWARDS

2026 **RISE-MICCAI 2026 Travel Grant**
One of only 5–10 awarded; supported attendance and presentation at MICCAI 2026, Strasbourg, France

LatinX in AI (LXAI) Travel Award
Awarded to attend and present at NeurIPS 2026, Sydney, Australia

High-Impact Mobility Fund (FMAI)
Universidad de los Andes – institutional co-financing for presenting high-impact papers at international conferences

2025 **1st Place – LISA 2025 Challenge (Tasks 2a & 2b)**
1st of 94 teams; Low-field Pediatric Brain MRI Segmentation Challenge, RISE-MICCAI, MICCAI 2025; earned selection for the RISE-MICCAI Paper-Lead Mentorship Program

2023 **Mitacs Globalink Research Internship Award**
Highly competitive (estimated acceptance under 5%); funded research internship at Dalhousie University, Canada

Silver Medal – iGEM Design League
Competed with an interdisciplinary systems & synthetic biology team of students from physics, biology, biomedical and environmental engineering, and mathematics

TEACHING EXPERIENCE

2026 – Present **Graduate Teaching Assistant** – Deep Learning & Computer Vision
M.Sc. in Artificial Intelligence (MaIA, online), Universidad de los Andes

- Lead the computational component of the course, teaching students to implement the algorithms behind the theory covered by the faculty, from classical classifiers to 3D reconstruction with Gaussian splatting.
- Manage course delivery and respond to student questions throughout each cohort.

2021 – 2024 **Physics Tutor**
Prisma Academic Support Center, Department of Physics, Universidad de los Andes
Tutored early-stage students one-to-one and in groups across introductory mechanics, thermodynamics, electrodynamics, waves and fluids, using an active-learning approach.

2022 – 2024 **Teaching Assistant**
Universidad de los Andes
Integral Calculus & Differential Equations (Honors) and Physics II – two semesters each; graded assessments, ran office hours, and supported course delivery.